Lesson-19

Limits Continuity Diff

Expansions:

i. $\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$

ii. $\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots$

iii. $\tan x = x + \frac{x^3}{3} + \frac{2x^5}{15} + \dots$

iv. $\sin^{-1} x : x + \frac{x^3}{3!} + \frac{9x^5}{5!} + \dots$

v. $\tan^{-1} x : x - \frac{x^3}{3} + \frac{x^5}{5} + \dots$

Indeterminants

$\frac{0}{0}, \frac{\infty}{\infty}, 1^{\infty}, \infty - \infty, 0^0, \infty^0, 0.\infty$

LH

formula

Common & Simplify

Convert to $\frac{0}{0}$ $\frac{\infty}{\infty}$

Rationalise

Factorise

(for Square root functions)

1^{∞}: $\lim_{x \to a} (1 + f(x))^{g(x)} = e^{\lim_{x \to a} f(x).g(x)}$

"Hint": Use $\cos D \pm \cos D$ According to question.

Trigonometry $\to$ Limits.

$1 + \cos 2x = 2\cos^2 x$

$1 - \cos 2x = 2\sin^2 x$

$\sin^2 x + \cos^2 x = 1$

i. $\lim_{x \to a} \frac{x^n - a^n}{x - a} = na^{n-1}$

ii. $\lim_{x \to 0} \frac{\sin px}{x} = p$ $\quad \lim_{x \to 0} \frac{\tan kx}{x} = k$

iii. $\lim_{x \to 0} \frac{1 - \cos 2x}{2x} = 2$

iv. $\lim_{x \to 0} \frac{\log(1 + px)}{x} = p$

v. $\lim_{x \to 0} (1 + x)^{\frac{1}{x}} = e$

vi. $\lim_{x \to 0} \frac{e^x - 1}{x} = 1$ $\quad$ vii. $\lim_{x \to 0} \frac{a^x - 1}{x} = \log a$ $\quad$ viii. $\lim_{x \to 0} \frac{\log(1 + kx)}{x} = k$

vi. $(1 + x)^p = 1 + px + \frac{p(p-1)}{2!} x^2 + \dots$

vii. $e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$

viii. $a^x = 1 + \frac{x \log a}{1!} + \frac{x^2 (\log a)^2}{2!} + \cdots$

(xi): $\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots$

(x): $\log(1-x) = -\left(x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{4} + \cdots\right)$

LH Derivable

$x = 0. = \lim_{h \to 0.} \frac{f(x) - f(0)}{(\ )}$

$LHL =, \neq RHL$

by

$[x^2], [x] \rightarrow$ check

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Polynomial Equation

like $2x^2 + x, x^2, x^3 \ldots$ is Alway continuous ...